Faculty of Science, Engineering and Built Environment

SEB101 Engineering Physics

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

The unit will commence with a calculus treatment of kinematics of motion in one, two and three dimensions, including parabolic motion. The application of Newton's laws to a wide variety of phenomena will be investigated and the concepts of work, energy and the conservation laws discussed. Rotational kinematics and dynamics will be introduced, including the concepts of moment of inertia and torque. Also introduced are electrical concepts of charge, field, voltage and circuits. Other topics to be covered include vectors, statics, centre of mass and oscillations.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: John Long
Campus: Geelong Waurn Ponds
Email: john.long@deakin.edu.au
Phone: +61 3 522 72896
Other members of the team and how to contact them

Details of any other unit team members will be advised through the unit site at the start of trimester.
Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Students appreciated the online support and the recorded lectures.
- Students also appreciated the active nature of the campus seminars.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- the unit content and practicals have been revised
- new videos on specific topics have been added

- improvements have been made to the equipment used in the lab experiments
- re-introduced in person lectures for campus students.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Explain basic principles in physical mechanics, dynamics, electric fields and circuits.	GLO1: Discipline-specific knowledge and capabilities
ULO2	Apply these principles to natural phenomena.	GLO1: Discipline-specific knowledge and capabilities
ULO3	Solve technical problems in basic mechanics, dynamics and electricity.	GLO4: Critical thinking GLO5: Problem solving
ULO4	Perform and report on basic physical measurements.	GLO2: Communication
ULO5	Employ experimental methodology.	GLO1: Discipline-specific knowledge and capabilities

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible for a pass in this unit, students must achieve a minimum of 40% in the End-of-Unit Assessment.

Brief summary of the hurdle requirements	Rationale
To be eligible to obtain a pass in this unit, students must achieve a minimum of 40% in the End-of-unit assessment.	In accordance with external accreditation requirements, the School of Engineering incorporates a hurdle requirement in order to achieve the Unit Learning Outcomes (ULOs) and Course Learning Outcomes (CLOs) of the course.

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	Laboratory write-up
Brief description of assessment task	There will be five experiments each requiring an individual laboratory write up. This is an individual assessment task.

Detail of student output	Traditionally, working scientists and engineers record their work in bound notebooks. You will do the same as practice in engineering note-keeping. The practical write-up is your diary entry of what you did in the experiment and what happened. It is best hand-written in a notebook. The notebook entry for each experiment has the same structure: 1. Title, date, and names of your lab 2. Aim of the experiment 3. Introduction and background. What you expect to see 4. Experimental details of what you did in the experiment, including any deviations from the procedure given in the lab 5. Results and calculations 6. Discussion of what the experiment means, uncertainties, and answers to the questions posed in the experiment. 7. Conclusion summarising the key results of the experiment 8. List of any literature consulted for the experiment Details on writing the lab notebook are contained in the introduction to the *SEB101 Lab Manual*. Note that these are not formal reports. They are a diary entries of what you did in the experiments and what results you obtained. Students should spend no more than five hours performing the experiment and writing it up in their notebooks.
Grading and weighting (% total mark for unit)	50% (5 x 10%) Marked
This task assesses your achievement of these Unit Learning Outcome(s)	ULO2: Apply these principles to natural phenomena. ULO4: Perform and report on basic physical measurements. ULO5: Employ experimental methodology. These ULOs are assessed by evaluating student progress, experimental results, calculations, professional behaviour and attention in the laboratory; and by the quality of the written submission as compared with the rubric criteria.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline knowledge and capabilities appropriate to the level of study related to a discipline or profession. GLO2: Communication: using oral, written and interpersonal communication to inform, motivate and effect change.
How and when you will receive feedback on your work	Student marks and feedback will be provided by means of a rubric posted to the SEB101 unit site. A sample lab write-up of a similar, but not identical, experiment will be posted to the unit website. Feedback for practicals 1 and 2 will normally be given by the next practical session. Feedback for practicals 3, 4 and 5 will be given during the End-of-unit assessment period.

When and how to submit your work	Scan your lab notebook and submit the scanned version to the drop box via the unit site. Use the Adobe Acrobat format. Each experiment is to be a separate file. Your name, student number, and the name of your lab partners must appear on the first page of each experiment. The filename must be of the form "*surname_experiment title_student No*.pdf". Individual photographs (JPEG or similar) of pages from your lab notebook are not acceptable. **Due dates:** 1. Practical 1: Monday @ 8pm (Week 4) 2. Practical 2: Monday @ 8pm (Week 7) 3. Practicals 3, 4 and 5: Monday @ 8pm (Week 12) Please refer to the unit site for further details. Any submissions that do not follow these instructions (including any reports submitted by email) will experience significant delays in marking. Note that students should spend no more than five hours performing a practical experiment and writing it up.

Summative assessment task 2

Title	**Problem solving tasks**
Brief description of assessment task	Written answers and solutions to questions/problems. This is an individual assessment task.
Detail of student output	Students are to submit written solutions to a series of assigned physics problems.
Grading and weighting (% total mark for unit)	20% (2 x 10%) Marked
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Explain basic principles in physical mechanics, dynamics, electric fields and circuits. ULO2: Apply these principles to natural phenomena. ULO3: Solve technical problems in basic mechanics, dynamics, and electricity. These ULOs are assessed by evaluating students' solutions and workings to a set of given problems against a set of standard solutions.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline knowledge and capabilities appropriate to the level of study related to a discipline or profession. GLO4: Critical thinking - evaluating information using critical and analytical thinking and judgment. GLO5: Problem solving - creating solutions to authentic (real world and ill-defined) problems.
How and when you will receive feedback on your work	Feedback and answers will be provided to the student approximately two weeks after the due date via the SEB101 unit site.
When and how to submit your work	Scan workings to an Acrobat .PDF file and submit it to the corresponding assignment folder on the unit site. The submission date of problem solving task 1 will be **Monday of Week 5, 31 March at 8pm (AEDT)** and for problem solving task 2 **Monday of Week 9, 5 May at 8pm (AEST).**

Summative assessment task End-of-unit assessment or Examination

Title	**End-of-Unit Assessment - Written**
Brief description of assessment task	An timed online assessment which will include a combination of multiple choice and short answer questions.

How you demonstrate your achievement	Students will be required to undertake a 2-hour written assessment. A written assessment is a form of assessment that asks you to use text to communicate your knowledge, skill or understanding of a topic. It is completed online. At a set release time, you will be given the assessment instructions and will complete the assessment by a set deadline.
Grading and weighting (% total mark for unit)	30%
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Explain basic principles in physical mechanics, dynamics, electric fields and circuits. ULO2: Apply these principles to natural phenomena. ULO3 Solve technical problems in basic mechanics, dynamics, and electricity. This assessment measures how well the student has understood physics concepts and can apply these basic problems on natural phenomena. These ULOs are assessed by evaluating students' solutions and workings to a set of given problems against a set of standard solutions.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: Discipline knowledge and capabilities: appropriate to the level of study related to a discipline or profession. GLO4: Critical thinking: evaluating information using critical and analytical thinking and judgment. GLO5: Problem solving: creating solutions to authentic (real world and ill-defined) problems. This assessment measures how well the student has understood physics concepts and can apply these to basic problems involving critical thinking.
How and when you will receive feedback on your work	Deakin University will release the final assessment results at the stipulated timeframe. Students will receive a mark, which is an indicator of their overall performance in this unit of study. Students can also make an appointment to get feedback on their assessment.
When and how to submit your work	Students will be required to undertake a timed online assessment via the Unit site during the End-of-unit assessment period. You will start and complete at any time within a scheduled 24-hour window, with a 2-hour time limit. Further details will be provided on the unit site. It is the responsibility of students to review their End-of-Unit Assessment timetable when it is released via StudentConnect. Further information including timetable release dates is available at end-of-unit assessments and examinations.

Formative assessment

(for practice and feedback rather than marks and grades)
In addition to the above, you will have a better chance of success in this Unit if you complete these tasks – you will get valuable experience and feedback.

Title	**Formative assessment - weekly homework exercises**
Brief description	This formative task is optional but highly recommended. Weekly homework exercises. Suggested exercises are given in the document posted to the Unit site, "Readings and suggested homework-problems."

This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Explain basic principles in physical mechanics, electricity and engineering moments ULO2: Apply these principles to natural phenomena ULO3: Solve technical problems in basic mechanics and electricity. This assessment measures how well the student has understood physics concepts and can apply these basic problems on natural phenomena. These ULOs are self-assessed by evaluating students' solutions and workings to a set of given problems against a set of standard solutions
When and how to submit your work	No due date. Physics problems are to be done in private study.
How and when you will receive feedback on your work	Feedback and answers will be provided to the student via the unit website.

Note: It is Engineering School's policy that in formative matters, students assess their own learning as much as possible.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

2 x 1 hour lectures per week, 1 x 2 hour seminar per week, 5 x 3 hour practical experience (laboratories) per trimester.

Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 2 x 1 hour online lectures per week, 1 x 1.5 hour seminar per week and 2 at home practicals and 3 on campus practicals per trimester. Students are required to attend and participate in practical experiences (laboratories) at the Waurn Ponds (Geelong) Campus during the trimester intensive activities as detailed in the unit site.

Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.

This will include educator guided online learning activities within the unit site.

SEB101 is taught according to the "cloud-learning" model. Teaching of primary concepts will be by means of the text material, the online study guide, and the associated cloud-concepts.

Except for the prescribed textbook by Halliday et.al, all unit learning material will be available online via the SEB101 unit site. Students should watch all the cloud-concept videos associated with the current topic.

Students, you are reminded that responsibility for your learning lies with you.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
Prescribed text(s): Halliday et al, 2020, Fundamentals of Physics ANZ Edition + Interactive Ebook Code, 1st Australia and New Zealand Ed, John Wiley & Sons Australia, Ltd.

The texts and reading list for SEB101 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

The prescribed textbook is essential reading, and is the single most important resource for SEB101. It should be brought to all lectures, seminars, and labs.

An electronic version of the textbook is available for purchase at:
https://www.wiley.com/en-au/Fundamentals+of+Physics%2C+1st+Australian+and+New+Zealand+Edition-p-9780730382867/.

Online learning resources

Each student must access the following online resources, obtained on the SEB101 unit site:

- the syllabus
- the lab manual
- online study guides
- embedded online *Cloud-Concept* videos.
- lecture and seminar notes

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Suitable reference books for this unit include:

- Knight, Randall, *Physics for Scientists and Engineers: a Strategic Approach with modern physics*, 5th Global Edition Pearson/Addison-Wesley, San Francisco, 2022.
- Giancoli, D.C., *Physics for Scientists & Engineers with modern physics*, 5th edition, Pearson, 2022.

Other resources you may find useful are:

- The Library has numerous books on general physics. Call No. 530 is where most are located.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic	Textbook references	Related Assessment	Assessment due date
1	3 March 2025	Introduction Kinematics **Practical assessment activity:** Pendulum and Hooke's Law	Halliday, chaps 1-2	Pracs, Assignment 1	
2#	10 March 2025	Vectors 2-D motion **Practical assessment activity:** Pendulum and Hooke's Law	Halliday, chaps 3-4	Pracs, Assignment 1	
3	17 March 2025	Newton's laws Applications of Newton's laws **Practical assessment activity:** Pendulum and Hooke's Law	Halliday, chaps 5-6	Pracs, Assignment 1	
4	24 March 2025	Work and energy Conservation of energy **Practical assessment activity:** Inclined Pane	Halliday, chaps 7-8	Pracs, Assignment 1, Assignment 2	**Assessment 1: Practical 1 write-up** due Monday 24 March
5	31 March 2025	Rotation **Practical assessment activity:** Inclined Pane	Halliday, chaps 10-11	Pracs, Assignment 2	**Assessment 2: Problem solving task 1** due 31 March
6	7 April 2025	Centre of mass Static equilibrium **Practical assessment activity:** Inclined Pane	Halliday, chaps. 9 and 12	Pracs, Assignment 2	
7*~	14 April 2025	Oscillations **Practical assessment activity:** Inclined Pane	Halliday, chap 15	Assignment 2	**Assessment 1: Practical 2 write-up** due Monday 14 April

8	28 April 2025	Electric charge Electric fields **Practical assessment activity:** Projectiles, Friction, Ohm's Law	Halliday, chaps 21-22	Pracs, Assignment 2	Intensive week
9	5 May 2025	Electric potential Capacitance **Practical assessment activity:** Projectiles, Friction, Ohm's Law	Halliday chaps 24-25	Pracs, Assignment 2	**Assessment 2: Problem solving task 2** due 5 May
10	12 May 2025	Current, resistance, Ohm's law **Practical assessment activity:** Projectiles, Friction, Ohm's Law	Halliday chap 26	Pracs	
11	19 May 2025	DC Circuits **Practical assessment activity:** Projectiles, Friction, Ohm's Law	Halliday chap. 27	Pracs	
12	26 May 2025				**Assessment 1: Practicals 3, 4 and 5 write-up** due Monday 26 May
13-14^	2 June 2025				**End-of-unit assessment**

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed